Total No. of Questions : 8]

SEAT No. :

PA-912

[Total No. of Pages : 3

[5927]-342

B.E. (Computer Engineering)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester - VII) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Solve the matrix chain multiplication for the following 6 matrix problem using Dynamic programming. **[10]**

Matrix	A_1	A_2	A_3	A_4	A_5	A_6
Dimensions	10×20	20×5	5×15	15×50	50×10	10×15

b) Explain Greedy strategy: Principle, control abstraction, time analysis of control abstraction with suitable example. **[8]**

OR

Q2) a) Explain the 'dynamic programming' approach for solving problems. Write a dynamic programming algorithm for creating an optimal binary search tree for a set of 'n' keys. Use the same algorithm to construct the optimal binary search tree for the following 4 keys. **[10]**

Key	A	B	C	D
Probability	0.1	0.2	0.4	0.3

b) Explain Dynamic programming: Principle, control abstraction, time analysis of control abstraction with suitable example. **[8]**

Q3) a) Explain the 'branch and bound' approach for solving problems. Write a branch and bound algorithm for solving the 0/1 Knapsack problem. Use the same algorithm to solve the following 0/1 Knapsack problem. The capacity of the Knapsack is 15 kg. **[9]**

Item	A	B	C	D
Profit (Rs.)	18	10	12	10
Weight (kg.)	9	4	6	2

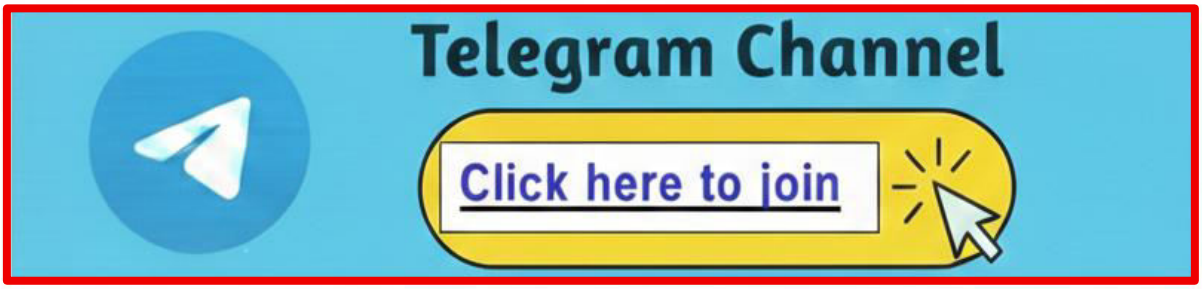
b) Explain with suitable example Backtracking. Principle, control abstraction, time analysis of control abstraction. **[8]**

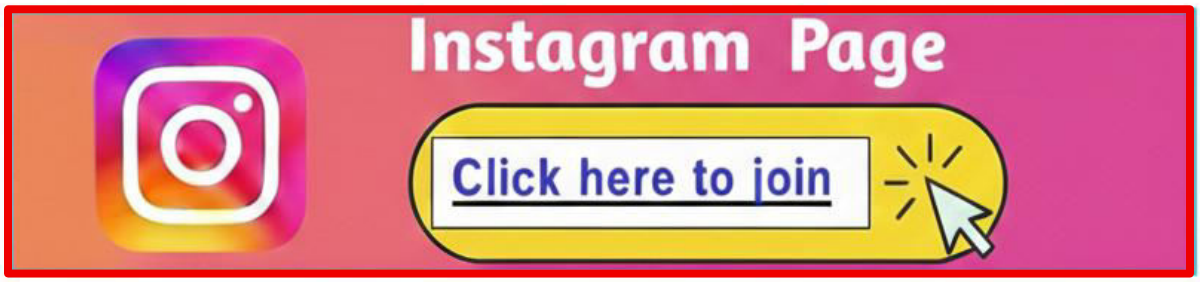
OR

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Q4) a) What is Branch and Bound method? Write control abstraction for Least Cost search? **[9]**

b) Explain the backtracking with graph coloring problem. Find solution for following graph **[8]**

	C_1	C_2	C_3	C_4	C_5
C_1	0	1	0	1	0
C_2	1	0	1	0	0
C_3	0	1	0	1	1
C_4	1	0	1	0	1
C_5	0	0	1	0	0

Adjacency matrix for graph G

Q5) a) Write short notes on the following. **[10]**

i) Aggregate Analysis

ii) Accounting Method

iii) Potential Function method

iv) Tractable and Non-tractable Problems

b) Write short notes on with suitable example of each **[8]**

i) Randomized algorithm

ii) Approximation algorithm

OR

Q6) a) What is Potential function method of amortized analysis? To illustrate Potential method, find amortized cost of PUSH, POP and MULTIPOP stack operations. **[9]**

b) What is embedded algorithm? Explain Embedded system scheduling using power optimized scheduling algorithm. **[9]**

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Q7) a) Write short notes on the following. **[10]**

i) Multithreaded matrix multiplication.

ii) Multithreaded merge sort

iii) Distributed breadth first search

iv) The Rabin-Karp algorithm

b) With respect to Multithreaded Algorithms explain Analyzing multithreaded algorithms, Parallel loops, Race conditions. **[7]**

OR

Q8) a) Write and explain pseudo code for Multi-threaded merge sort algorithm. How parallel merging gives a significant parallelism advantage over Merge Sort? **[9]**

b) Write a pseudo code for naïve string matching algorithm and Rabin-Karp algorithm for string matching and analyze the same. **[8]**

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Total No. of Questions : 8]

SEAT No. :

P3147

[Total No. of Pages : 3

[6004]-480A

B.E. (Computer Engineering)

DESIGN ANALYSIS OF ALGORITHMS

(2019 Pattern) (Semester - VII) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Consider the following instance of the knapsack problem. Find the optimal solution by using dynamic programming approach. **[10]**

Item	Weight	Profit
1	2	$12
2	1	$10
3	3	$20
4	2	$15

Capacity of the knapsack = 5.

b) What is job scheduling algorithm? How job scheduling algorithm can be solved using Greedy algorithmic approach? Explain your answer with respect to Principle, control abstraction, time analysis of control abstraction, of greedy approach for the following instance of knapsack problem. **[8]**

Each job is associated with a deadline and profit.

Job	J_1	J_2	J_3	J_4	J_5
Deadline	2	1	3	2	1
Profit	60	100	20	40	20

OR

Q2) a) What is greedy approach? Explain Job scheduling algorithm using Greedy approach for following examples. Give the sequence of job scheduling. **[8]**

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Input: Four jobs with following deadlines and profits

Job1D	deadline	Profit
a	4	20
b	1	10
c	1	40
d	1	30

Input: Five Jobs with following deadlines and profits

Job1D	Deadline	Profit
a	2	100
b	1	19
c	2	27
d	1	25
e	3	15

b) What is optimal binary search tree? How dynamic programming approach is used to build OBST for following tale. **[10]**

	1	2	3	4
Keys→	10	20	30	40
Frequency→	4	2	6	3

Q3) a) Explain with suitable example Backtracking: Principle, control abstraction, time analysis of control abstraction. **[8]**

b) Compare between greedy method and dynamic programming with respect to. **[9]**

i) Feasibility.

ii) Optimality.

iii) Recursion.

iv) Memorization.

v) Time complexity.

OR

Q4) a) What is sum of subset problem? Solve sum of subset problem for following instance using backtracking approach. **[8]**

Input : set [] = {2, 3, 5, 6, 8, 10}, sum = 10

b) What is Branch and Bound method? Write control abstraction for Least cost search? **[9]**

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Q5) a) What is amortized analysis? Explain aggregate and potential function methods used for amortized analysis with respect to stack operations? **[9]**

b) What is potential function method, of amortized analysis? To illustrate potential method, find amortized cost of PUSH, POP and MULTIPOP stack operations. **[9]**

OR

Q6) a) Write short notes on the following. **[10]**

i) Aggregate analysis.

ii) Accounting Analysis.

iii) Potential function method.

iv) Tractable and Non-tractable problems.

b) Write short notes on with suitable example of each. **[8]**

i) Randomized algorithm.

ii) Approximation algorithm.

Q7) a) Write and explain pseudo code for multi-threaded merge sort algorithm. How parallel merging gives a significant parallelism advantage over merge Sort? **[9]**

b) i) Explain an algorithm for Distributed Minimum Spanning Tree. **[8]**

ii) Write and explain Rabin-Karp algorithm for string matching.

OR

Q8) a) Write short notes on the following. **[10]**

i) Multithreaded matrix multiplication.

ii) Multithreaded merge sort.

iii) Distributed breadth first search.

iv) The Rabin-Karp algorithm.

b) With respect to Multithreaded Algorithms explain Analyzing multithreaded algorithms, Parallel loops, Race conditions. **[7]**

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Total No. of Questions : 8]

SEAT No. :

P-6551

[Total No. of Pages : 3

[6181]-101
B.E. (Computer Engineering)
DESIGN AND ANALYSIS OF ALGORITHM
(2019 Pattern) (Semester-I) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Write High-level description of job sequencing algorithm. Let number of jobs (n)=5; Profit vector P={20, 15, 10, 5, 1); Deadline vector D={2, 2, 1, 3, 3) Find the feasible solutions. What is the optimal solution and maximum profit? **[9]**

b) Consider the following instance of the knapsack problem. Find the optimal solution by using dynamic programming approach. **[9]**

Item	Weight	Profit
1	2	$12
2	1	$10
3	3	$20
4	2	$15

Capacity of the knapsack = 5.

OR

Q2) a) What is Job scheduling algorithm? How job scheduling algorithm can be solved using Greedy algorithmic approach? Explain your answer with respect to Principle, control abstraction, time analysis of control abstraction, of greedy approach for the following instance of Knapsack problem. **[12]**

Each job is associated with a deadline and profit.

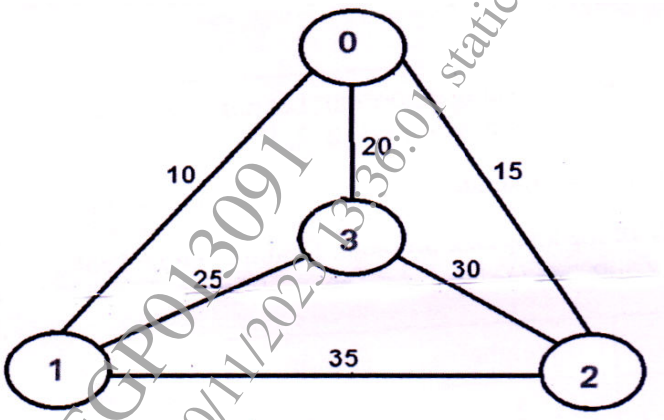
Job	J_1	J_2	J_3	J_4	J_5
Deadline	2	1	3	2	1
Profit	60	100	20	40	20

b) Write steps for Greedy approach for Job sequencing. **[6]**

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Q3) a) What is branch and bound algorithmic strategy? Apply branch n bound algorithmic strategy to solve traveling salesman problem for **[9]**



b) Explain with suitable example Backtracking: Principle, control abstraction, time analysis of control abstraction. **[8]**

OR

Q4) a) Explain the 'branch and bound' approach for solving problems. Write a branch and bound algorithm for solving the 0/1 Knapsack problem. Use the same algorithm to solve the following 0/1 Knapsack problem. The capacity of the knapsack is 15 kg. **[9]**

Item	A	B	C	D
Profit(Rs.)	18	10	12	10
Weight (Kg.)	9	4	6	2

b) What is sum of subset problem? Solve sum of subset problem for following instance using backtracking approach **[8]**

Input: set[ ] = {2, 3, 5, 6, 8, 10}, sum = 10

Q5) a) What is amortized analysis? Explain the aggregate method with example. **[9]**

b) What is Potential function method of amortized analysis? To illustrate Potential method, find amortized cost of PUSH, POP and MULTIPOP stack operations. **[9]**

OR

Q6) a) What are special needs of embedded algorithm? Which sorting algorithm is best for embedded systems? Why? **[6]**

b) Explain Randomized and Approximate algorithms. **[4]**

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c) What is randomized algorithm? Give any example of randomized algorithm? Also explain Random variable, Binomial random variable and-Mathematics for Randomized algorithm. **[8]**

Q7) a) i) Explain an algorithm for Distributed Minimum Spanning Tree.

ii) Write and explain Rabin-Karp algorithm for string matching. **[10]**

b) With respect to Multithreaded Algorithms explain Analyzing multithreaded algorithms, Parallel loops, Race conditions. **[7]**

OR

Q8) a) Write and explain pseudo code for Multi-threaded merge sort algorithm. How parallel merging gives a significant parallelism advantage over Merge Sort? **[9]**

b) For string matching, working module q = 11, how many spurious hits does the Rabin- Karp matcher encounters in Text T = 31415926535.....**[8]**

i) T = 31415926535

ii) P=26

iii) Here T.Length = 11 so Q = 11

iv) And P mod Q = 26 mod 11 = 4

v) Now find the exact match of P mod Q...

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Total No. of Questions : 8]

SEAT No. :

PB2243

[Total No. of Pages :3

[6263]-81

B.E. (Computer Engineering)

DESIGN AND ANALYSIS OF ALGORITHMS

(2019 Pattern) (Semester- VII) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Assume suitable data if necessary.***
4) ***Figures to the right indicate maximum marks.***

Q1) a) Write a control abstraction for greedy method. Comment on the time complexity of this abstraction? **[6]**

b) Find an optimal solution for the following knapsack instance using greedy method. Number of objects n = 5. Capacity of knapsack m = 100. **[8]**

Objects	Weight	Profit
01	20	10
02	30	20
03	66	30
04	40	40
05	60	50

c) Comment on the statement "Problem which does not satisfy the principle of optimality cannot be solved by dynamic programming". **[4]**

OR

Q2) a) Write a control abstraction for dynamic programming strategy. Comment on the time complexity of this abstraction? **[8]**

b) Consider 4 matrices A1, A2, A3 and A4. The orders of these matrices are given below: **[10]**

Matrix	Order
A1	3×5
A2	5×4
A3	4×2
A4	2×4

Find the optimal sequence of chain matrix multiplication of these matrices using dynamic programming approach. Clearly give the final sequence and total number of multiplications involved.

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Q3) a) Assume that a graph with n vertices is represented by an adjacency matrix G. Let there be "m" number of colours available. Write a recursive backtracking algorithm to colour all the vertices of the graph. What is the time complexity of this algorithm? **[8]**

b) Consider three items along with respective weights and value as **[9]**

	Weight	Value	Value/Weight
O1	5	6	6/5 = 1.2
O2	4	5	5/4 = 1.25
O3	3	4	4/3 = 1.3

Assume the Knapsack capacity m = 7. Solve this 0/1 Knapsack problem using LC branch and bound method.

OR

Q4) a) Compare backtracking with branch and bound method with respect to: search technique, exploration of state space tree and king of problems that can be solved. **[8]**

b) Consider set A of five numbers {5,10,15,20.25}. We wish to find the subset of A such that sum of the numbers in this subject is equal to 30. Solve this problem to find the first solution using backtracking approach. Show space tree being created. **[9]**

Q5) a) What are randomized algorithms? Enlist and explain in brief the primary reasons for using randomized algorithms. **[8]**

b) What are approximation algorithms? Based on the approximation ratio, classify the approximation algorithms. **[9]**

OR

Q6) a) Explain the methods of amortized analysis. Give suitable example. **[8]**

b) Suppose you are working on an embedded system for a medical device that monitors patient vital signs. The device continuously collects data from various sensors and needs to process and display this information in real-time. The data includes timestamps, temperature readings, heart rate, and blood pressure measurements. Suggest suitable sorting algorithm for this scenario. Clearly justify your answer with respect to key factors. **[9]**

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Q7) a) Write a Rabin-Karp string matching algorithm. Input to the algorithm be: Original text "t" of length n and pattern text being matched is "p" of length m. What is the expected runtime and worst-case runtime of this algorithm? **[10]**

b) Write multi-threaded merge sort algorithm. Briefly discuss how does it differ from conventional merge sort. **[8]**

OR

Q8) a) Consider the graph represented by an adjacency matrix: **[10]**

	A	B	C	D	E	F	G
A	0	1	1	0	0	0	0
B	1	0	0	1	1	0	0
C	1	0	0	0	0	1	1
D	0	1	0	0	0	0	0
E	0	1	0	0	0	0	0
F	0	0	1	0	0	0	0
G	0	0	1	0	0	0	0

Show stepwise process how the distributed breadth first search algorithm works on the above graph.

b) What do you understand by spawn and sync keywords used in multithreaded programming? Explain with the help of suitable example. **[8]**



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Total No. of Questions : 8]

SEAT No. :

PC-2368

[Total No. of Pages : 4

[6354]-485
B.E. (Computer Engineering)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester - VII) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Assume suitable data if necessary.*
4) *Figures to the right indicate full marks.*

Q1) a) You are given a set of tasks, each with a deadline and a penalty for missing the deadline. The objective is to schedule these tasks in a way that minimizes the total penalty incurred. However, you can only work on one task at a time, and once a task is started, it must be completed before moving on to the next task. Additionally, you can't start a task after its deadline has passed. Design a greedy algorithm to efficiently schedule these tasks to minimize the total penalty and prove its correctness. **[8]**

b) Suppose we have a knapsack with a maximum weight capacity of 15 units, and we have the following items with their respective weights (Wi) and values (Vi) :

Objects	Weight	Value
O1	8	10
O2	6	8
O3	4	3
O4	2	4

Use greedy approach to maximize the total value of items we can put into the knapsack without exceeding its weight capacity. **[8]**

c) With respect to dynamic programming, what do you understand by optimal substructure? **[2]**

OR

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Q2) a) We are given the sequence {4, 10,3, 12,20, and 7}. We are given with five matrices of the size 4 × 10, 10 × 3,3 × 12, 12 × 20,20 × 7 respectively. Use dynamic programming to solve chain matrix multiplication. **[10]**

b) Under what situation do you think the dynamic programming approach for solving a knapsack problem might struggle to find the optimal solution? Briefly explain. **[4]**

c) Enlist the uses of writing control abstraction for any algorithmic strategies. **[4]**

Q3) a) Consider a graph represented by the adjacency matrix given below :

	A	B	C	D	E	F	G
A	0	1	1	0	0	0	0
B	1	0	0	1	1	0	0
C	1	0	0	0	0	1	1
D	0	1	0	0	0	0	0
E	0	1	0	0	0	0	0
F	0	0	1	0	0	0	0
G	0	0	1	0	0	0	0

Use a recursive backtracking algorithm to colour this graph with three colours R,G,B **[8]**

b) Consider three items along with respective weights and value as

	Weight	Value
O_1	10	12
O_2	8	10
O_3	6	8

Assume the Knapsack capacity m = 14. Solve this 0/1 Knapsack problem using LC branch and bound method. **[9]**

OR

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Q4) a) We have a salesman who needs to visit four cities (A, B, C, D) and return to the starting city. The distances between these cities are as follows :

Distance from A to B: 10 units

Distance from A to C: 15 units

Distance from A to D: 20 units

Distance from B to C: 35 units

Distance from B to D: 25 units

Distance from C to D: 30 units

Find the shortest possible route that visits each city exactly once and returns to the starting city. Use branch and bound method to find the optimum route for traveling salesman, assume A as a starting point of the tour. **[8]**

b) Write a short note on LC branch and bound method. **[5]**

c) What are the drawbacks of branch and bound method? **[4]**

Q5) a) What are the advantages and disadvantages of : **[8]**

i) Aggregate Analysis

ii) Accounting Method

b) What are approximation algorithms? Based on the approximation ratio, classify the approximation algorithms. **[9]**

OR

Q6) a) Why potential function method cannot be used for analysing binary counter? Explain **[8]**

b) Comment on the following statements : **[9]**

i) "The knapsack problem is NP-hard"

ii) "Boolean Satisfiability Problem (SAT) is NP-complete"

iii) "Minimum spanning tree is tractable problem"

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Q7) a) Write a Rabin-Karp string matching algorithm. Let input to the algorithm be Original text "t" of length n and pattern text being matched is "p" of length m. What is the expected runtime and worst–case runtime of this algorithm? **[10]**

b) Briefly explain performance measures – speedup, efficiency, throughput, contention, and latency of rnultithreaded algorithms. **[8]**

OR

Q8) a) Consider the graph represented by an adjacency matrix : **[10]**

	A	B	C	D	E	F	G
A	0	1	1	0	0	0	0
B	1	0	0	1	1	0	0
C	1	0	0	0	0	1	1
D	0	1	0	0	0	0	0
E	0	1	0	0	0	0	0
F	0	0	1	0	0	0	0
G	0	0	1	0	0	0	0

Show stepwise process how the distributed breadth first search algorithm works on the above graph.

b) If we have two matrices of the order m x n and n x p then what will be the time complexity of multiplying these matrices in conventional approach and in multithreaded approach. Discuss. **[8]**



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Total No. of Questions : 8]

SEAT No. :

PD4576

[Total No. of Pages : 3

[6404]-81
B.E.(Computer Engineering)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester - VII) (410241)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Assume suitable data, if necessary.***
4) ***Figures to the right indicate full marks.***

Q1) a) Give a mathematical formulation for: **[8]**

i) Fractional Knapsack problem

ii) 0/1 Knapsack problem.

b) Consider a set of seven activities. The start and finish time of these activities are given in the following table. **[9]**

Activity	Start time	Finish time
A	1	3
B	3	4
C	2	5
D	0	7
E	5	9
F	8	10
G	11	12

Use the greedy algorithmic strategy to compute an execution schedule having maximum number of non-conflicting activities.

OR

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Q2) a) With respect to dynamic programming, what is the principle of optimality? Give a mathematical representation for the same. **[7]**

b) Consider the 0/1 Knapsack problem with the total weight capacity of 7. The weights and associated values (profits) of 4 items are given below. Use dynamic programming approach to find the optimal solution to this problem. **[10]**

Item	Weight	Value(Profit)
A	1	1
B	3	4
C	4	5
D	5	7

Q3) a) Consider a graph, which is represented by the adjacency matrix given below: **[10]**

	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	1
D	0	1	1	0

We wish to colour this graph using Red and Black colours using a backtracking algorithm. Show the process of colouring it in stepwise manner using state space diagram.

b) Prove that the full state space tree of finding sum of subset of n elements using backtracking will have (excluding the leaf nodes)2^n-1 nodes. **[4]**

c) What are the major drawbacks of branch and bound method? **[4]**

OR

Q4) a) State the sum of subsets problem. Write an algorithm to solve this problem using backtracking method. Comment on the time complexity of this algorithm. **[8]**

b) Consider the three objects. The weights and associated values are given below.

	weight	value
O_1	5	6
O_2	4	5
O_3	3	4

Assume the Knapsack capacity m = 7. Solve this 0/1 Knapsack problem using LC branch and bound method. **[10]**

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Q5) a) Give an amortized analysis of a k bit binary counter using aggregate method. **[6]**

b) What are tractable and non-tractable problems? Give examples. **[7]**

c) Does randomized algorithm for quick sort, improves the average case time complexity? **[4]**

OR

Q6) a) Explain with the help of example the methods of amortized analysis. **[6]**

b) What is an approximation algorithm? How does performance ratios useful in approximation algorithms? **[7]**

c) What are randomized algorithms? Enlist few reasons to use randomized algorithms. **[4]**

Q7) a) Write a simple multithreaded matrix multiplication algorithm based on parallelizing relevant loops of the conventional procedure. **[8]**

b) Explain in brief race condition in multithreaded algorithms. **[4]**

c) What do you understand by *spawn* and *sync* keywords used in multithreaded programming? **[6]**

OR

Q8) a) Write distributed breadth search algorithm. What is its advantage over the conventional approach? **[8]**

b) Write a Rabin-Karp string matching algorithm. Input to the algorithm be: Original text "t" of length n and pattern text being matched is "p" of length m. What is the expected runtime and worst-case runtime of this algorithm? **[10]**

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Total No. of Questions : 8]

SEAT No. :

PE2192

[Total No. of Pages : 3

[6584]-91
B.E. (Computer Engineering)
DESIGN AND ANALYSIS OF ALGORITHMS
(2019 Pattern) (Semester - VII) (410241)

Time :2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8*
2) *Neat diagrams must be drawn whenever necessary.*
3) *Assume suitable data, if necessary.*
4) *Figures to the right indicate full marks.*

Q1) a) Compare greedy algorithmic strategy and dynamic programming strategy. **[6]**

b) We are given a set of 4 jobs. The deadline and the associated profit with each job is given below. **[8]**

Jobs	A	B	C	D
Deadline	2	1	1	2
Profit	75	40	50	35

Assume that every job takes a single unit of time. Maximize the total profit if only one job can be scheduled at a time using a greedy algorithmic strategy.

c) Comment on the statement "Greedy method always provides the optimal solution". **[3]**

OR

Q2) a) Write a control abstraction for dynamic programming strategy. Comment on the time complexity of this abstraction? **[7]**

b) Find the binomial coefficients of $(x+y)^5$, using dynamic programming approach. **[10]**

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Q3) a) Consider the set A of five numbers {5,10,15,20,25}. We wish to find the subset of A such that sum of the numbers in this subset is equal to 30. Solve this problem to find the first solution using backtracking approach. Show space tree being created. **[10]**

b) Write a control abstraction for branch and bound strategy, Enlist few applications of branch and bound method. **[8]**

OR

Q4) a) Traveling salesperson has to make trip between three cities P,Q,R. The cost matrix is given below. No edge is indicated by ∞. The tour will start from city P. Solve this travelling salesperson problem using branch and bound method. **[10]**

G=	P	Q	R
P	∞	4	2
Q	3	∞	4
R	1	8	∞

b) Write an algorithm for graph colouring problem using backtracking method. Let the graph consists of n nodes and that there are m-number of availble colours. What is the time complexity of this algorithm? **[8]**

Q5) a) Prove that amortized cost per operation in k-bit binary counter is O(1).**[5]**

b) What are intractable problems? Does traveling salesperson Problem belong to this class of problems? Justify your answer. **[6]**

c) Briefly explain any two methods of amortized analysis. **[6]**

OR

Q6) a) Explain in brief amortized analysis. Consider stack as an example and single push, single pop and multiple pop operations. Give the amortized costs for these stack operations. **[9]**

b) Does randomized algorithm for quick sort, improves the average case time complexity? Discuss. **[4]**

c) Comment on the statement "The problem of Searching an ordered list or searching an unordered list belongs to the class of tractable problems".**[4]**

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Q7) a) Let the hash code for the text be "4133124" and that for the pattern matching text be "124". Use modulo arithmetic with the prime number 13 and apply Rabin-Karp string matching algorithm for this case. Show the stepwise process of matching the pattern. What is the time complexity of this algorithm? **[10]**

b) Compare and contrast row-wise and block-wise approaches in multithreaded matrix multiplication with respect to: division of work, communication overhead and cache efficiency. **[8]**

OR

Q8) a) Consider the text T and the pattern text P. Write a naïve string-matching algorithm. Show the stepwise process of working of this algorithm if T = "KITCHEN" and P = "HEN". **[10]**

b) Write a distributed algorithm to find the minimum spanning tree. **[8]**

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